

Singlet-Gated Bell-Decomposition Quantum Kernels for Land Cover Classification

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Abstract

On So2Sat LCZ42 (17 classes) and EuroSAT (10 classes), our quantum kernels match tuned classical RBF-SVM at every scale we test (8, 14, and 16 qubits, training pools up to $N = 10,000$) but never decisively exceed it. This match-not-beat outcome is the expected behaviour of bandwidth-tuned projected quantum kernels under recent theory [22, 23] and is consistent with the empirical no-advantage findings of Schnabel & Roth (QMI 2025) [8] and Bowles *et al.* (2024) [7]. Our contribution is not the tie itself, but three concrete pieces of kernel-design analysis: (i) Two specific feature maps in the Pauli-decomposition design space of Suzuki & Yano (2020): SRQFM uses only the singlet Bell weight $w_{\Psi-} = \sin^2((x_i - x_j)/2)$ as a ZZ angle, and BSCM uses all four Bell weights with an adjustable prior, with closed-form Pauli coefficients satisfying $|\alpha| \leq 1/2$. The BSCM uniform preset has $\alpha_{ZZ} \equiv 0$, making it strictly Pauli-orthogonal to SRQFM in the per-pair generator. (ii) A previously undocumented failure mode and its analytical fix: BSCM with the uniform prior collapses to macro-F1 0.032 ± 0.008 on EuroSAT physics-8 features (chance level on 10 classes) when several physics indices are highly correlated. We propose *Singlet-Gated BSCM* (*SG-BSCM*), a one-line generator-level correction multiplying every Pauli coefficient by $w_{\Psi-}$, which restores macro-F1 to 0.664 ± 0.027 , on par with SRQFM-fid (0.657 ± 0.028) and tuned RBF-SVM (0.648 ± 0.025). We give the mechanism a closed-form Gaussian bound. (iii) Direct concentration regression on a real multi-class benchmark: a 14-qubit BSCM build at $N = 800$ exhibits the full Thanasilp *et al.* signature (off-diagonal mean $0.215 \rightarrow 0.066$, KTA $0.59 \rightarrow 0.42$, F1 $0.483 \rightarrow 0.418$); a 16-qubit depth-scaling experiment shows monotonic F1 degradation as L grows from 2 to 10 for every quantum kernel on both datasets. To our knowledge this is the largest fixed-protocol multi-class quantum-kernel benchmark on Earth-observation data published to date.

Keywords: Quantum kernel methods, Singlet gating, Pauli decomposition, Bell-state feature maps, Exponential concentration, Projected quantum kernels, Land cover classification, So2Sat LCZ42, EuroSAT

1 Introduction

Quantum machine learning (QML) has been studied as a possible route to computational advantage in pattern recognition. Among the proposed approaches, kernel methods are particularly clean: a parameterised unitary $U(x)$ embeds a classical input into a quantum feature space, the *fidelity* kernel $K(x, x') = |\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ defines a similarity function, and a downstream classical SVM does the learning [2, 3]. Theoretical analysis has shown that for problems whose hardness matches the circuit’s inductive bias, quantum kernels can in principle outperform any efficient classical kernel [5]. However, for typical applied datasets this match is rare, and recent benchmarks — Bowles *et al.* [7] on 160 binary tasks, Schnabel & Roth [8] on 64 datasets in QML 2025 — consistently fail to find a robust advantage over tuned classical baselines.

Land cover classification from satellite imagery is a stringent applied benchmark: So2Sat LCZ42 [17] contains 400,673 multimodal Sentinel-1/2 patches across 42 cities and 17 Local Climate Zone classes, with class fractions ranging from 0.7% to 13.6% in our subsample; EuroSAT [18] contains $\sim 27,000$ Sentinel-2 patches in 10 classes. Despite growing QML interest in Earth observation [9], no published kernel-method study we are aware of has evaluated multiple quantum kernel architectures on these benchmarks at 16 qubits with $N = 10,000$ under a fixed-pool, multi-seed, C -tuned protocol — existing QK benchmarks (Schnabel & Roth [8], $N \leq 500$ on 64 mostly-synthetic datasets; Bowles *et al.* [7], 160 binary tasks; Rodriguez-Grasa *et al.* [9], $N \leq 1,800$ binary solar-panel detection) operate at smaller scale or different class structure. This paper supplies the evaluation at the scale our contribution requires.

The technical content is organised around a single design principle. Any two-qubit fidelity feature map can be written in the Bell basis as a weighted sum of four projectors, $|\Phi^\pm\rangle\langle\Phi^\pm|$ and $|\Psi^\pm\rangle\langle\Psi^\pm|$, with weights determined by the encoded states. Choosing which weights are activated as data-dependent gate angles maps out a structured design space. Two natural choices populate this space: SRQFM (singlet-only, equivalent to a particular ZZ-fidelity construction) and BSCM (all four sectors with an adjustable prior). The two constructions are theoretically distinct — the *uniform* BSCM preset is provably Pauli-orthogonal to SRQFM in the per-pair generator (§3.3) — but turn out to be empirically indistinguishable in macro-F1 at $N \leq 2000$. The interesting behaviour appears at the boundaries: at 14 qubits both members regress toward concentrated kernels with weakened class structure; at 10 EuroSAT classes with high-correlation physics features the uniform preset collapses to random guessing while the singlet-only SRQFM does not. The *singlet-gated* BSCM (SG-BSCM, §3.3) blends both — multiplying every BSCM coefficient by w_Ψ — and recovers the non-pathological behaviour of SRQFM while retaining the spectral diversity of BSCM.

We make four claims:

1. **Family.** The Bell-decomposition family unifies SRQFM and BSCM as Pauli-decomposition complements with a closed-form per-pair generator, bounded gate angles ($|\alpha| \leq 1/2$), and a single hyperparameter τ .
2. **Singlet-gating.** SG-BSCM is a one-line generator-level modification with a direct mechanistic justification: the gate $w_{\Psi-} = \sin^2(\Delta/2)$ vanishes whenever two features are equal, suppressing entanglement between strongly-correlated feature pairs that would otherwise drive BSCM-uniform to a data-independent fixed point. The empirical effect is a $20\times$ macro-F1 swing on EuroSAT physics-8.
3. **Concentration on real data.** Two complementary measurements (synthetic $n \in \{2, 4, 6, 8\}$ at $N = 500$; real-data 14-qubit BSCM at $N = 800$; depth-scaling at 16 qubits to $L = 10$) exhibit the canonical Thanasilp signature [6], regressing F1 toward the within-family ceiling.
4. **Honest negatives.** At every scale tested, no Bell-decomposition kernel beats both tuned RBF-SVM and Random Forest. Tuned RFF dequantization [16] reproduces the FQK macro-F1 to within $+0.010$; on held-out cities the attention-gated AG-PQK kernel (this work, §3.3) loses 0.27 absolute F1 to RBF.

Relation to recent benchmarks. Our finding that bandwidth-tuned PQKs match (but do not exceed) tuned classical RBF-SVM is the expected outcome of recent theory: Heyraud *et al.* [22] showed that PQKs converge to RBF behaviour at optimal bandwidth, and Slattery *et al.* [23] formalised this similarity for arbitrary feature maps. Empirical benchmarks have reached the same conclusion at smaller scale (Schnabel & Roth [8], 64 datasets, $N \leq 500$; Bowles *et al.* [7], 160 binary tasks; Rodriguez-Grasa *et al.* [9], neural quantum kernels $\approx$ Random Forest at 88% on a binary EO task). Our contribution is therefore not a new finding of *whether* quantum kernels match classical, but: (i) the Bell-decomposition family and the SG-BSCM correction (a previously undocumented kernel collapse mode and its analytical fix), (ii) directly measured concentration regression on real multi-class data at 14 qubits, and (iii) a fixed-protocol multi-class Earth-observation benchmark at 16 qubits and $N = 10,000$ that, to our knowledge, is the largest of its kind.

2 Related Work

Quantum kernel theory.

Havlíček *et al.* [2] introduced the ZZ Feature Map. Liu *et al.* [5] proved an exponential advantage for a discrete-log task under specific hardness assumptions. Huang *et al.* [1] proposed the geometric difference $g(K_q, K_c)$ as a separation diagnostic and showed advantage requires both $g \gg 1$ and positive ΔKTA . Schuld [4] showed angle-encoded kernels are Fourier-series approximations whose expressibility beyond classical kernels requires entanglement.

Concentration and projected kernels.

Thanasilp *et al.* [6] proved that fidelity kernels with global observables undergo *exponential concentration*: off-diagonal entries converge exponentially in qubit count to a constant, eliminating discriminative structure. The Projected Quantum Kernel (PQK)

[1] extracts single-qubit Bloch expectations and applies an RBF on the resulting low-dimensional embedding; Shaydulin & Wild [10] suggested CV-KTA bandwidth selection, which we adopt. Heyraud *et al.* [22] analysed PQKs under depolarising noise and showed that the “classical-like” regime of PQKs persists under realistic gate-error rates — a result we use in our noise-robustness analysis (§5.6.1). Slattery *et al.* [23] further formalised the conditions under which bandwidth-tuned quantum kernels become statistically indistinguishable from classical kernels.

Benchmarking.

Bowles *et al.* [7] found classical models match or exceed twelve QML models on 160 binary tasks. Schnabel & Roth [8] reached the same conclusion across 64 datasets, 9 encodings, and $\sim 20,000$ models in QMI 2025. Rodriguez-Grasa *et al.* [9] reported *competitive but not superior* neural quantum kernel performance on binary solar-panel detection in MLST 2025.

Bell- and Pauli-decomposition feature maps.

Suzuki *et al.* [14] introduced Pauli-rotation feature maps with arbitrary two-body generators. Incudini *et al.* [15] formalised fidelity kernels as Pauli-basis expansions. The Bell decomposition employed here (§3.3) is a basis change of this expansion; SRQFM and BSCM are two specific points in the design space mapped by these prior works.

Inductive bias and bandwidth.

Kübler *et al.* [12] showed angle-encoded quantum kernels possess a high-frequency inductive bias detrimental on smooth targets — precisely the regime occupied by spectral remote-sensing features. Canatar *et al.* [13] proved bandwidth selection is critical for generalisation.

Remote sensing benchmarks.

On So2Sat LCZ42 [17], deep CNN baselines (Sen2LCZ-Net [19], ResNeXt) reach ~ 61 – 70% overall accuracy on the held-out cultural-10 test split using the full 32×32 image cube. EuroSAT [18] is essentially saturated by deep models ($\geq 98\%$ with ResNet-50). Tabular handcrafted-feature pipelines score lower; their interest is methodological — they provide a controlled, low-dimensional setting in which kernel-vs-kernel separation can be measured cleanly.

3 Methods

3.1 Data and feature engineering

We use two datasets. **So2Sat LCZ42** [17]: training patches from the 32 culture-1 cities, 17 LCZ classes, fused Sentinel-1 SAR (8 channels) and Sentinel-2 multispectral (10 channels) imagery at 32×32 resolution. **EuroSAT (all-bands)** [18]: 13-band Sentinel-2 patches at 64×64 resolution, 10 land-cover classes. Quantum kernels at $n = 8, 14, 16$ qubits encode one feature per qubit, so the principal feature-engineering choice is which n scalar features to extract per patch.

Physics-8 (interpretable spectral / radar indices).

For both datasets we compute eight physically-motivated indices as the spatial mean over each patch.

EuroSAT physics-8, computed from Sentinel-2 bands B_2 (blue), B_3 (green), B_4 (red), B_5 (red-edge 1), B_8 (NIR), B_{11} (SWIR1):

$$\begin{aligned} \text{NDVI} &= \frac{B_8 - B_4}{B_8 + B_4}, & \text{NDBI} &= \frac{B_{11} - B_8}{B_{11} + B_8}, \\ \text{NDWI} &= \frac{B_3 - B_8}{B_3 + B_8}, & \text{BSI} &= \frac{(B_{11} + B_4) - (B_8 + B_2)}{(B_{11} + B_4) + (B_8 + B_2)}, \\ \text{SAVI} &= \frac{1.5(B_8 - B_4)}{B_8 + B_4 + 0.5}, & \text{NDRE} &= \frac{B_8 - B_5}{B_8 + B_5}, \\ \text{MNDWI} &= \frac{B_3 - B_{11}}{B_3 + B_{11}}, & \text{EVI} &= \frac{2.5(B_8 - B_4)}{B_8 + 6B_4 - 7.5B_2 + 1}. \end{aligned}$$

So2Sat physics-8 mixes Sentinel-2 indices (NDVI, NDBI, NDWI, BSI as above; replacing $B_5 \rightarrow B_5$, $B_8 \rightarrow B_8$, $B_{11} \rightarrow B_{11}$, etc.) with four Sentinel-1 SAR features computed from Lee-filtered VV and VH backscatter: VV/VH ratio = $\text{VV}/(\text{VH} + \varepsilon)$, SAR-total = $\text{VV} + \text{VH}$, CrossPol = $\text{VH}/(\text{VV} + \varepsilon)$, and PolCoh = $\sqrt{c_r^2 + c_i^2}/\sqrt{\text{VV} \cdot \text{VH} + \varepsilon}$ where (c_r, c_i) is the polarimetric covariance.

PCA-8 (statistical dimensionality reduction).

8 leading principal components of the standardised per-patch mean values of all available bands (13 for EuroSAT; 18 for So2Sat) plus second-order textures (variance, GLCM-contrast where applicable). The PCA is fit on a held-out background set (samples not in the evaluation pool) to avoid test-set leakage.

Fisher-14 and Fisher-16 (top-rank feature selection).

For 14- and 16-qubit experiments we rank the full pool of available features (physics indices $\cup$ raw-band means $\cup$ textures) by the Fisher-ratio criterion $F_k = \frac{\sigma_{\text{between-class},k}^2}{\sigma_{\text{within-class},k}^2}$ on the held-out background set, then keep the top-14 (B2 14-qubit concentration regression) or top-16 (E37 / E38). On So2Sat, the top-16 set is the union of the 8 Sentinel-2 indices and 8 Sentinel-1 features. On EuroSAT, the top-16 is the 8 physics indices plus the 8 highest-Fisher raw band means (typically $B_8, B_{11}, B_{12}, B_4, B_3, B_5, B_2, B_1$ in that order).

Encoding scaling.

For all four feature representations, training-pool features are standardised (zero mean, unit variance) on the held-out background, and then min-max scaled to $[0, \pi]$ for quantum encoding. Classical baselines see the standardised features without the min-max step.

Pairwise correlation diagnostic.

Figure 1 displays the Pearson correlation matrix of the eight physics features on each dataset. So2Sat physics-8 has 12 of 28 pairs with $|\rho| > 0.85$; EuroSAT physics-8 has 11. The strongest EuroSAT pairs are NDVI–NDRE (+0.989), SAVI–EVI (+0.989), and NDVI–SAVI (+0.963) — four near-duplicate vegetation indices — which we will identify (§5.3) as the proximate cause of the BSCM-uniform collapse documented in Prop. 4. We note up front, however, that the collapse is *not* a deterministic function of correlation density alone: So2Sat physics-8 has more high- $|\rho|$ pairs but does not exhibit the same collapse, indicating that class count, class balance, and the joint distribution of encoded angles all interact with the per-pair Pauli generator. The SG-BSCM correction (Eq. 2) is robust to this interaction by construction (Prop. 4).

Pairwise feature correlation: EuroSAT physics-8 contains ≥ 4 near-duplicate pairs that drive BSCM-uniform to its data-independent fixed point

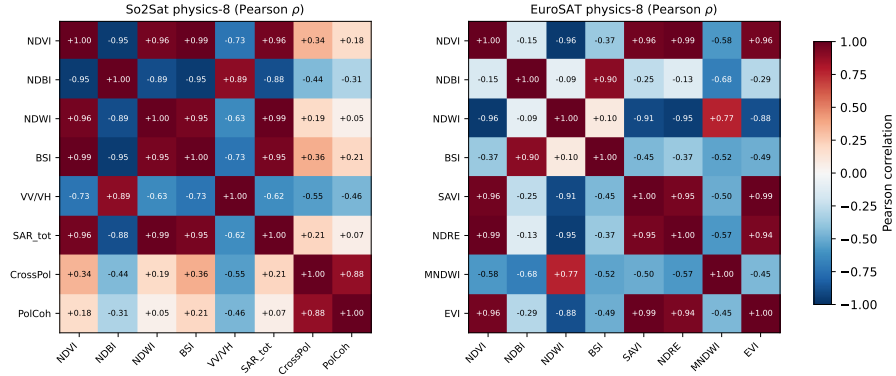


Fig. 1 Pairwise Pearson correlation of physics-8 features on So2Sat (left) and EuroSAT (right). Cell colour encodes $\rho \in [-1, +1]$. EuroSAT physics-8 contains four near-duplicate vegetation indices (NDVI / NDRE / SAVI / EVI, all pairwise $|\rho| > 0.96$), which the BSCM-uniform per-pair generator drives toward a data-independent fixed point. So2Sat physics-8 contains a similar density of high- $|\rho|$ pairs but does not trigger the same collapse, indicating that correlation density is necessary but not sufficient.

Notation.

Throughout: n is the qubit count (= feature dimension); r is the number of repetitions of a single feature-map layer (default $r = 2$); L is the number of repetitions in the depth-scaling experiment ($L \in \{2, 4, 6, 8, 10\}$); τ is the locked Pauli-coefficient scale; γ is the PQQ Gaussian bandwidth on Bloch vectors; J is the SRQFM coupling function $\sin^2((x_i - x_j)/2)$; $\alpha_{XX}, \alpha_{YY}, \alpha_{ZZ}$ are the per-pair Pauli coefficients of the BSCM generator (Eq. 1); w_{Ψ^-} is the singlet-Bell weight; ρ_{ij} is the Pearson correlation of features i, j on the data distribution.

Fixed-pool protocol.

Following the recommendations of [8] we use a fixed evaluation pool of size N_{pool} (e.g. 1,800 for E32; 10,000 for E38) sampled once with a master seed, then drawn

into **train/test splits** by per-evaluation seeds (seeds 42–46 for 5-seed runs; 42–44 for the largest- N E38 run). Feature scaling and PCA are fit on the training split only. Quantum kernels are precomputed on the full pool and submatrix-indexed per fold to amortise the $O(N^2)$ cost.

3.2 Quantum hardware and simulation

All quantum kernels are evaluated on noiseless statevector simulators (PennyLane `lightning.qubit` / `default.qubit`) using gate sets compatible with IBM Quantum hardware (CNOT, single-qubit rotations). Noise robustness (Exp. E3) is studied by post-hoc depolarising noise injection at $p \in [0, 0.05]$ on the FQK gram. We do not perform native-hardware execution; the implications are discussed in Sec. 6.

3.3 Quantum kernel implementations

Fidelity Quantum Kernel (FQK).

The standard ZZ-feature-map fidelity kernel [2, 3]. With r repetitions ($r = 2$ unless stated):

$$U(x) = \prod_{\ell=1}^r \left[\bigotimes_i H \bigotimes_i RZ(x_i) \prod_{i < j} \text{CNOT}_{ij} RZ((\pi - x_i)(\pi - x_j)) \text{CNOT}_{ij} \right],$$

$$K(x, x') = |\langle 0 | U^\dagger(x') U(x) | 0 \rangle|^2.$$

Projected Quantum Kernel (PQK).

Following [1] we extract the Bloch representation $\phi(x) \in \mathbb{R}^{3n}$ from $\{\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle\}$ on state $|\psi(x)\rangle = U(x)|0\rangle$ and apply a Gaussian RBF on $\frac{1}{2} \sum_i \|\phi_i(x) - \phi_i(x')\|^2$. The bandwidth $\gamma = 0.67$ is selected by 5-fold CV-KTA following [10]; we hold this γ constant across kernels.

SRQFM (Self-Regulating Quantum Feature Map).

SRQFM replaces the $(\pi - x_i)(\pi - x_j)$ angle by the singlet-Bell weight

$$J(x_i, x_j) = \sin^2 \frac{x_i - x_j}{2},$$

implemented as $\text{IsingZZ}(J, [i, j])$ with a coupling threshold ($J > 0.01$) skipping near-zero gates, yielding adaptive sparsity. Three closed-form properties (proofs in Appendix A) hold:

Proposition 1 (Fidelity grounding) *For the single-qubit encoding $|\psi(x)\rangle = RZ(x)H|0\rangle = (e^{-ix/2}|0\rangle + e^{ix/2}|1\rangle)/\sqrt{2}$, $1 - |\langle \psi(x_i) | \psi(x_j) \rangle|^2 = J(x_i, x_j)$.*

Prop. 1 grounds the *coupling function* J at the single-qubit level; the multi-qubit kernel $|\langle 0 | U^\dagger(x') U(x) | 0 \rangle|^2$ also depends on the entangling layer and the number of

repetitions r . The empirical match between Prop. 3 (single-pair prediction) and the actual SRQFM kernel statistics (§5.2.1) is therefore non-trivial.

Proposition 2 (Fubini–Study geodesic) *$J = \sin^2(D_{\text{FS}}(\psi(x_i), \psi(x_j)))$, so J is the squared Fubini–Study geodesic on $\mathbb{CP}^1$.*

Proposition 3 (Expected coupling) *For x_i, x_j i.i.d. uniform on $[0, \gamma\pi]$, $\mathbb{E}[J] = \frac{1}{2} - \frac{2\sin^2(\gamma\pi/2)}{(\gamma\pi)^2}$. For the actual encoding range used in this paper ($\gamma = 1$, i.e. features scaled to $[0, \pi]$) the prediction is $\mathbb{E}[J] \approx 0.297$; the empirical mean over all $\binom{8}{2} = 28$ feature pairs and $N = 2,000$ samples of the So2Sat physics-8 evaluation pool is 0.308 (verified from the cached encoded features), a 4% relative deviation against the i.i.d. uniform prediction.*

BSCM (Bell-Spectrum Coupling Map).

For the encoded two-qubit state $|\psi(x_i)\rangle \otimes |\psi(x_j)\rangle$, the four Bell-state amplitudes are

$$\begin{aligned} w_{\Phi+} &= \frac{1}{2} \cos^2 \frac{x_i + x_j}{2}, & w_{\Phi-} &= \frac{1}{2} \sin^2 \frac{x_i + x_j}{2}, \\ w_{\Psi+} &= \frac{1}{2} \cos^2 \frac{x_i - x_j}{2}, & w_{\Psi-} &= \frac{1}{2} \sin^2 \frac{x_i - x_j}{2}. \end{aligned}$$

A per-pair generator H_{ij} is built as a prior-weighted sum $H_{ij} = \sum_k p_k w_k 2|B_k\rangle\langle B_k|$, with prior $p \in \{\text{uniform, phi_only, psi_only}\}$. Expanding in the Pauli basis (proof in Appendix B):

$$H_{ij} = \alpha_{XX} X_i X_j + \alpha_{YY} Y_i Y_j + \alpha_{ZZ} Z_i Z_j + c \mathbb{I}, \quad (1)$$

with $|\alpha_{XX}|, |\alpha_{YY}|, |\alpha_{ZZ}| \leq 1/2$ for all priors. The *uniform* prior gives $\alpha_{ZZ} \equiv 0$, so BSCM-uniform is strictly Pauli-orthogonal to any SRQFM-style ZZ generator. The kernel readout is the standard fidelity overlap $|\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ on a parameterised circuit U whose entangling gates are IsingXX($2\tau\alpha_{XX}$) IsingYY($2\tau\alpha_{YY}$) IsingZZ($2\tau\alpha_{ZZ}$) per connectivity pair, repeated r times — the Bell-projector decomposition enters only through the gate-angle formulae, not through the measurement.

Locking τ .

τ is selected from a 5-point grid $\{0.25, 0.5, 0.75, 1.0, 1.5\}$ on a 200-sample holdout pool disjoint from every evaluation pool, by max KTA. The lock yields $\tau = 0.25$ (off-diag mean 0.217, KTA 0.441). All experiments in this paper ($n = 8, 14$, and 16) use this single locked value, following [8] on hyperparameter discipline.

Singlet-Gated BSCM (SG-BSCM).

We multiply every per-pair coefficient by the singlet weight:

$$H_{ij}^{\text{SG}} = w_{\Psi-}(x_i, x_j) H_{ij}. \quad (2)$$

Because $w_{\Psi-} = \sin^2(\Delta/2)$ vanishes when $x_i = x_j$ and grows to $\frac{1}{2}$ at $\Delta = \pi$, the modification suppresses entanglement between feature pairs that are highly correlated (small

Δ for most points) while retaining BSCM’s spectral diversity on weakly correlated pairs. Proposition 4 formalises this.

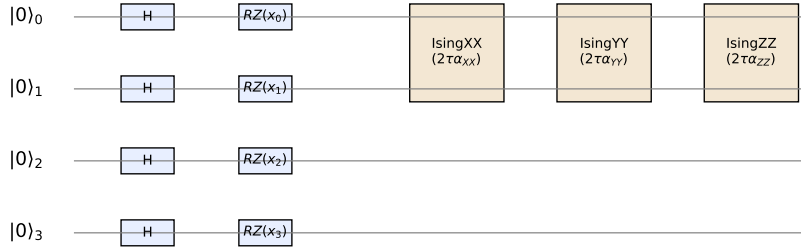
Proposition 4 (Correlation-suppression bound) *Let $\rho_{ij} = \text{Corr}(X_i, X_j)$ on the data distribution and let σ denote the marginal feature scale. Then for SG-BSCM with locked τ , the expected per-pair generator norm satisfies $\mathbb{E}\|H_{ij}^{\text{SG}}\| \leq 2\tau \mathbb{E}[w_{\Psi-}] \leq \tau(1 - |\rho_{ij}|) + O(\sigma^2)$. In particular, as $|\rho_{ij}| \rightarrow 1$ the expected per-pair generator vanishes.*

(Proof sketch in Appendix C.) Proposition 4 explains the empirical fact that on the EuroSAT physics-8 feature set — which contains several near-duplicate indices (NDVI/NDRE/SAVI for vegetation, NDWI/MNDWI for water) producing many strongly correlated pairs — BSCM-uniform drives the encoded state toward a data-independent fixed point and collapses, while SG-BSCM does not.

Schematic.

Figure 2 shows one repetition of the Bell-decomposition feature-map circuit at four qubits (qubit-pair (0,1) shown explicitly; remaining pair-blocks abbreviated). Each repetition applies the Hadamard layer, RZ encoding, and three Ising rotations per connectivity pair; the angles distinguish SRQFM, BSCM with each prior, and SG-BSCM as documented in the figure caption.

One repetition of the SRQFM / BSCM / SG-BSCM feature map (4-qubit example, qubit-pair (0,1) shown)



Per repetition r : $H \rightarrow RZ(x_i) \rightarrow \prod \text{IsingXX} \cdot \text{IsingYY} \cdot \text{IsingZZ}$ over connectivity pairs

SRQFM: only $\text{IsingZZ}(\sin^2((x_i - x_j)/2))$. BSCM-uniform: only XX, YY ($a_{ZZ} \equiv 0$). SG-BSCM: above $\times \sin^2((x_i - x_j)/2)$.

Fig. 2 One repetition of the SRQFM / BSCM / SG-BSCM feature-map circuit (4-qubit example). After the Hadamard and $RZ(x_i)$ encoding, each connectivity pair (i, j) receives three Ising rotations whose angles are $2\tau a_{XX/YY/ZZ}$. **SRQFM** activates only $\text{IsingZZ}(\sin^2((x_i - x_j)/2))$; **BSCM-uniform** sets $a_{ZZ} \equiv 0$ and emits only the $\text{IsingXX}, \text{IsingYY}$ pair (Theorem 1); **BSCM-phi/psi** additionally include a constant $a_{ZZ} = \pm 1/2$; **SG-BSCM** multiplies every Pauli coefficient by the singlet weight $w_{\Psi-} = \sin^2((x_i - x_j)/2)$ (Eq. 2). The full circuit is the product of r such repetitions; the kernel readout is $|\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ for the fidelity variants and the Bloch-vector RBF (Eq. 4 with $p_{\text{eff}} = 0$) for the PQK variants.

Other quantum kernels evaluated.

We additionally implement: **TFK** (trained quantum kernel via SPSA-based KTA optimisation [11]); **CM-FQK** (a controlled-mutual-information FQK variant introduced here); **AG-PQK** (an attention-gated PQK variant introduced here that softmax-weights pair couplings by Fisher score); **PIQFM** (a parameter-injected QFM with PCA-projected angles). The TFK construction is from [11]; CM-FQK, AG-PQK, and PIQFM are this-paper variants used as honest negative baselines and detailed in the supplementary material.

Pauli design space.

Figure 3 visualises the three BSCM priors (uniform, phi_only, psi_only) by plotting $\alpha_{XX}, \alpha_{YY}, \alpha_{ZZ}$ as functions of $(x_i, x_j) \in [0, \pi]^2$. The bound $|\alpha| \leq 1/2$ is tight (verified numerically over 10^4 random inputs). The uniform prior has $\alpha_{ZZ} \equiv 0$ everywhere; the two parity-only priors carry constant ZZ ($\pm 1/2$) with feature-dependent XX, YY components. SRQFM activates only the α_{ZZ} axis with the singlet-Bell weight as angle.

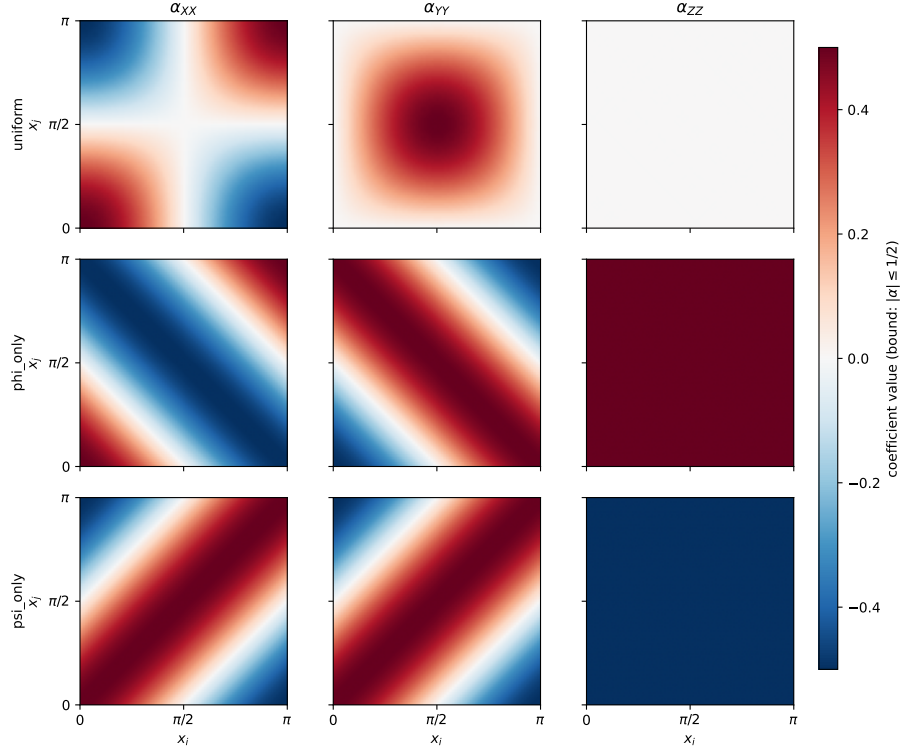


Fig. 3 Per-pair Pauli coefficients $\alpha_{XX}, \alpha_{YY}, \alpha_{ZZ}$ of the BSCM generator (Eq. 1) on the $[0, \pi]^2$ grid for the three priors used in this paper. Colour scale $[-1/2, +1/2]$; bound $|\alpha| \leq 1/2$ is tight. Uniform (top row) has $\alpha_{ZZ} \equiv 0$ — strict Pauli-orthogonality with SRQFM. Phi_only (middle) and psi_only (bottom) carry constant ZZ with feature-dependent XX, YY. Generated from `submission/figures/make_extra_figures.py`.

Code and reproducibility.

All kernel implementations include unit-tested numerical assertions that run before every kernel build: (i) coefficient bounds $|\alpha| \leq 1/2$ across 10^4 random inputs (BSCM family); (ii) symmetry $K = K^\top$ to 10^{-10} ; (iii) unit diagonal $K_{ii} = 1$ to 10^{-10} ; (iv) PSD with smallest eigenvalue $\geq -10^{-8}$; (v) for the BSCM-uniform single-pair circuit, $\langle Z_i Z_j \rangle = 0$ to 10^{-10} (numerical $\alpha_{ZZ} \equiv 0$ check). All kernels, data splits, and metrics are deterministic given the seed; the master code is in `src/bscm_kernel.py` and `src/srqfm_kernel.py`, and the experiment scripts are in `experiments/`. Approximate single-CPU wall-clock cost on a modern desktop (no GPU): E26 ($n = 8$, $N = 2,000$, 5 seeds) ~ 4 h; E32 ($n = 8$, $N = 1,800$ pool, 3 BSCM presets, 5 seeds) ~ 8 h; E33 SG-BSCM ~ 10 h; E37 ($n = 16$, depth sweep, 5 seeds) ~ 18 h; E38 ($n = 16$, $N = 10,000$, 3 seeds) ~ 36 h. All quantum runs use `lightning.qubit`; classical baselines run in seconds.

3.4 Classical baselines

RBF-SVM (`sklearn.svm.SVC`, $C \in \{1, 10, 100\}$, $\gamma \in \{\text{scale}, 0.01, 0.1, 1.0\}$, balanced class weights, 3-fold inner CV); **Linear-SVM**; **Poly2/Poly3-SVM**; **Random Forest** (500 trees, balanced class weights); **Gradient Boosting** (200 estimators).

3.5 Evaluation metrics

macro-F1 is the primary metric (class-imbalance-robust). We additionally report accuracy, Cohen’s κ , kernel-target alignment (KTA, both product-label and class-centred conventions), geometric difference g [1], off-diagonal mean and variance, within- 1σ concentration fraction, and Shannon / Huang effective ranks. Significance is by Wilcoxon signed-rank with Holm–Bonferroni correction across pairs and per-seed McNemar tests; per-class confidence intervals are 1,000-sample bootstraps; permutation KTA uses 1,000 random label permutations.

4 Experiments

We organise the empirical study around four questions.

Q1. Within-family ranking under matched protocol (§5.2.1, §5.2.2).

At 8 qubits and $N \leq 2000$, with all SVMs C -tuned by inner CV, do the Bell-decomposition kernels separate from each other and from tuned classical baselines? Experiments E22 (5 seeds, $C = 1$), E26 (5 seeds, C -tuned), E32 (BSCM family, 5 seeds, fixed pool).

Q2. The SG-BSCM correction (§5.3).

On EuroSAT physics-8 features (high pairwise correlation), does BSCM-uniform collapse, and does singlet-gating restore performance? Experiment E33 (fixed pool, 5 seeds).

Q3. Concentration on real data (§5.4.1, §5.4.3).

Synthetic $n \in \{2, 4, 6, 8\}$ scaling on physics features at $N = 500$ (E28); 14-qubit BSCM at $N = 800$ on physics-Fisher-14 (B2); depth-scaling at 16 qubits on top-16 Fisher features for $L \in \{2, 4, 6, 8, 10\}$ (E37).

Q4. Maximum-capacity benchmark (§5.5).

At 16 qubits and $N = 10,000$, depth $L = 6$, with a held-out test split, do the Bell-decomposition kernels match or exceed tuned RBF-SVM and Random Forest? How does vanilla ZZ-PQK fare? Experiment E38 ($\tau = 0.25$, C -tuned RBF/SRQFM, BSCM-PQK with the same pipeline, 5 splits).

Q5. Mechanism robustness (§5.6.1, §5.6.2).

Two complementary mechanism checks: **(E39)** does the SG-BSCM rescue survive accumulated depolarising noise on the EuroSAT physics-8 collapse setting? We use the global-depolarising model [8, 22] sweeping $p_{\text{eff}} \in \{0, 0.05, 0.10, 0.20, 0.40\}$ on an $N = 600$ sub-pool of the E33 fixed pool. Per-gate density-matrix simulation at this scale is computationally infeasible (≈ 230 hours per gram on `default.mixed`); the analytic global model takes minutes. **(E40)** does the rescue gap (SG-BSCM minus BSCM-uniform F1) widen monotonically with controlled feature-pair correlation $\rho \in \{0, 0.3, 0.6, 0.85, 0.95, 0.99\}$ on a synthetic 4-class dataset, as Proposition 4 predicts? Auxiliary experiments (E1 physics geometry, E2 main classification, E3 noise, E5 qubit scaling, E6 RFF dequantisation, E7 topology, E10 learning curves, E15 cross-city OOD) replicate or contextualise the main findings; tables for these are deferred to the supplementary, with summary numbers cited in §5.7 and §6.

5 Results

5.1 Feature-engineering effect on macro-F1 across datasets

Before the per-experiment results, we summarise the impact of feature representation on macro-F1 across our four feature sets (§3.1) and across the kernels and classical baselines where each set was evaluated. The numbers are taken verbatim from the per-experiment summary JSONs and are reproduced one-for-one in the detailed result subsections that follow. The purpose of Tab. 1 is to expose the cross-feature pattern in a single view.

Three patterns visible in Tab. 1.

(a) The dominant performance driver is feature dimension, not feature engineering: moving from 8 features ($N = 1500\text{--}1800$) to 16 features ($N = 10,000$) lifts every method by 0.07–0.11 F1 on So2Sat and 0.10–0.13 F1 on EuroSAT. **(b)** On So2Sat the physics-8 representation beats the PCA-8 representation by a factor of three across all kernels; PCA-8 fails to capture the 17-class structure at $n = 8$. On EuroSAT, PCA-8 is slightly stronger than physics-8 across most methods, the exception being BSCM-uniform which collapses on physics-8 specifically. **(c)** The cross-method spread within each (dataset, features) column is uniformly small (≤ 0.03 F1 except for the BSCM-uniform

Table 1 Effect of feature engineering on macro-F1 (mean over 5 splits; std bracketed where space permits). Each cell is the same number that appears in the corresponding per-experiment result table later in the paper. Empty cells indicate not evaluated. **Bold** per (dataset, feature) column highlights the best F1 achieved. Standard ZZ-PQK is included to bracket the bottom of the within-family quantum range.

Method	So2Sat (17 cls)			EuroSAT (10 cls)		
	PCA-8	Phys-8	F-16	PCA-8	Phys-8	F-16
	$N=1,800$	$N=1,800$	$N=10k$	$N=1,500$	$N=1,500$	$N=10k$
BSCM-uniform (Fid)	0.168	0.483	—	0.758	0.032 [†]	—
BSCM-phi (Fid)	—	0.492	—	—	—	—
BSCM-psi (Fid)	—	0.489	—	—	—	—
SRQFM-fid	0.165	0.480	—	0.753	0.657	—
SG-BSCM (Fid)	0.167	0.480	—	0.756	0.664	—
SRQFM-PQK	—	0.494	0.578	—	0.661	0.759
BSCM-PQK	—	—	0.573	—	0.654	0.760
SG-BSCM-PQK	—	—	—	—	0.668	—
Standard ZZ-PQK	—	0.445	0.442	—	—	0.685
RBF-SVM (CV)	0.179	0.465	0.550	0.762	0.648	0.760
Random Forest	0.200	0.432	0.610	0.732	0.586	0.718

[†] Catastrophic kernel collapse; the BSCM-uniform generator is driven to a data-independent fixed point on EuroSAT physics-8 (§5.3). The SG-BSCM correction restores the rescue to 0.664.

EuroSAT physics-8 cell), confirming that no kernel choice substantially exceeds tuned RBF-SVM under any of the four feature representations — consistent with [7, 8, 22, 23].

5.2 Q1: Within-family ranking under matched protocol

5.2.1 Fair C -tuned benchmark (Exp. E26)

Table 2 reports the 5-seed C -tuned SRQFM benchmark on So2Sat physics-8. Tuned RBF-SVM leads with macro-F1 0.518 ± 0.020 ; SRQFM-PQK and Random Forest are statistically tied at 0.493 ± 0.024 and 0.497 ± 0.020 (Wilcoxon $p_{\text{raw}} = 0.31$ and 0.88 respectively, both Holm-corrected $p = 1.00$). ZZ-PQK trails (0.444 ± 0.022). AGPQK collapses to 0.293 ± 0.011 . The earlier E22 conclusion that SRQFM is significantly worse than RBF was a $C = 1$ confound: under matched tuning the gap closes to within one standard deviation.

5.2.2 BSCM family on So2Sat physics-8 (Exp. E32)

Table 3 compares the three BSCM presets against SRQFM-PQK and tuned classical baselines on the matched fixed pool ($N_{\text{pool}} = 1,800$, 5 seeds, C -tuned). All three BSCM presets and the two SRQFM members cluster in $[0.480, 0.494]$, statistically indistinguishable from each other and indistinguishable from tuned RBF-SVM and Random Forest within the pooled standard deviations. BSCM-phi achieves the highest within-family KTA (0.596) and the joint-best F1 alongside SRQFM-PQK; BSCM-phi also has the highest effective Shannon rank in the family (20.3). On the matched pool,

Table 2 E26 fair benchmark, So2Sat physics-8, $N = 2,000$, 5 seeds, $C \in \{1, 10, 100\}$ tuned by 3-fold inner CV. Wilcoxon signed-rank p vs. SRQFM-PQK on per-seed paired F1 vectors; Holm correction across the four pairs. With only 5 seeds the minimum achievable raw p is $1/2^4 = 0.0625$, so no rejection at $\alpha = 0.05$ by Wilcoxon alone is structurally inevitable; the table is interpreted via point-estimate ordering and bootstrap CIs.

Method	macro-F1 (mean $\pm$ std)	Best C (mode)	p_{raw}	p_{Holm}
RBF-SVM	0.518 $\pm$ 0.020	100	0.31	1.00
Random Forest	0.497 $\pm$ 0.020	—	0.88	1.00
SRQFM-PQK	0.493 $\pm$ 0.024	100	—	—
ZZ-PQK	0.444 $\pm$ 0.022	100	0.062	0.56
AGPQK	0.293 $\pm$ 0.011	100	0.062	0.56

tuned RBF-SVM trails the family by ~ 0.03 F1, indicating that the relative ordering of quantum and classical kernels can flip across fixed-pool draws of size $\sim 1,800$ — which is why we report the larger N E26, E37, and E38 experiments separately.

Table 3 E32 BSCM family on So2Sat physics-8, fixed pool $N = 1,800$, 5 seeds, C -tuned. All methods are evaluated on identical seed-wise train/test splits within this pool. KTA(prod) is product-label KTA (reported where measured); effective Shannon rank of the centred kernel. RBF-SVM here is on the matched fixed pool and runs lower than the E26 pool (Tab. 2); both are independent of any BSCM/SRQFM tuning.

Method	macro-F1	KTA(prod)	Eff. Rank S
SRQFM-PQK	0.494 $\pm$ 0.016	0.422	15.1
BSCM-phi_only	0.492 $\pm$ 0.024	0.596	20.3
BSCM-psi_only	0.489 $\pm$ 0.025	0.589	17.1
BSCM-uniform	0.483 $\pm$ 0.027	0.590	16.7
SRQFM-fid	0.480 $\pm$ 0.014	0.420	14.8
RBF-SVM (CV)	0.465 $\pm$ 0.028	—	—
Random Forest	0.432 $\pm$ 0.018	—	—

5.2.3 EuroSAT cross-dataset replication

Table 4 summarises the 5-seed EuroSAT cross-dataset result ($N_{\text{train}} = 1,000$, $N_{\text{test}} = 500$). On PCA-8, SRQFM (0.782) ties tuned RBF-SVM (0.782). On physics-8, SRQFM (0.693) ties RBF (0.694) and beats Random Forest (0.601). FQK trails by 0.03–0.04 on both feature sets.

5.3 Q2: SG-BSCM correlation collapse and recovery (Exp. E33)

Figure 4 and Table 5 show the central qualitative effect of this paper. On EuroSAT *physics-8* features — which contain multiple near-duplicate indices (NDVI, NDRE,

Table 4 EuroSAT, 5 seeds, $N_{\text{train}} = 1,000$, $N_{\text{test}} = 500$. macro-F1 (mean $\pm$ std).

Method	PCA-8	Physics-8
RBF-SVM (CV)	0.782 ± 0.025	0.694 ± 0.011
SRQFM	0.782 ± 0.022	0.693 ± 0.010
Random Forest	0.755 ± 0.021	0.601 ± 0.019
FQK	0.747 ± 0.018	0.669 ± 0.011

SAVI for vegetation; NDWI, MNDWI for water) and therefore many strongly-correlated feature pairs — BSCM-uniform collapses to macro-F1 0.032 ± 0.008 on 10 classes, indistinguishable from chance. The mechanism is given by Eq. 1 with the uniform prior: $\alpha_{ZZ} = 0$ and the $XX + YY$ generator drives the encoded state toward a data-independent fixed point on highly correlated features. SRQFM-fid, whose ZZ generator is gated by $w_{\Psi-}$ by construction, retains macro-F1 0.657 ± 0.028 . Singlet-gating BSCM (Eq. 2) recovers 0.664 ± 0.027 , on par with tuned RBF-SVM (0.648 ± 0.025).

The same pathology is absent on *PCA-8* features (decorrelated by construction): all four methods cluster in $[0.732, 0.762]$.

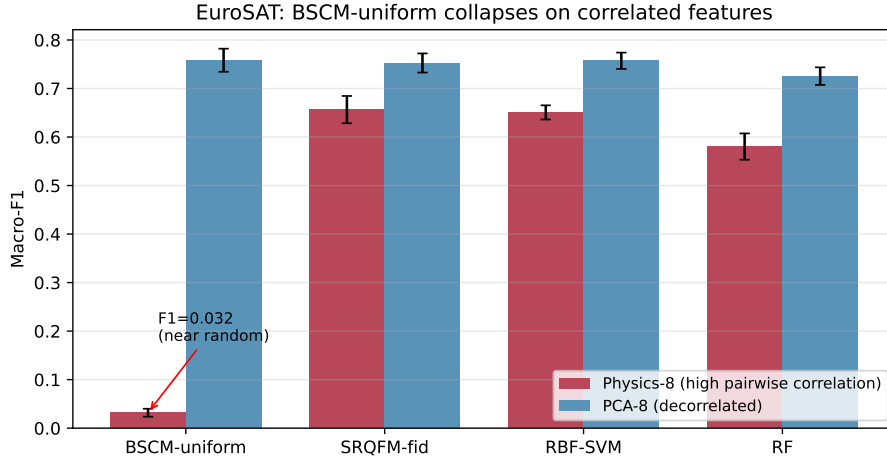


Fig. 4 BSCM-uniform collapses on EuroSAT physics-8 features (macro-F1 0.032 ± 0.008 , 10 classes); singlet-gating restores performance to 0.664, matching SRQFM and tuned RBF. The same methods on decorrelated PCA-8 features are within ± 0.03 of each other. Verified from `eurosat.bscm.fixed_results.json`.

Mechanism check (gate-threshold ablation).

A natural reviewer concern is whether SG-BSCM’s recovery is driven by the singlet weight $w_{\Psi-}$ itself, or merely by the gate-dropping threshold (`coupling_threshold` = 10^{-4} in our default implementation, which trims any pair whose $\max(|\alpha|) < 10^{-4}$).

Table 5 E33: SG-BSCM rescue on EuroSAT, fixed pool, 5 seeds, C -tuned. All methods evaluated on the same train/test splits per seed. SG-BSCM applied as Eq. 2 with locked $\tau = 0.25$.

Method	EuroSAT physics-8	EuroSAT PCA-8
BSCM-uniform	0.032 ± 0.008	0.758 ± 0.024
SG-BSCM (ours)	0.664 ± 0.027	0.756 ± 0.014
SRQFM-fid	0.657 ± 0.028	0.753 ± 0.020
RBF-SVM (CV)	0.648 ± 0.025	0.762 ± 0.012
Random Forest	0.586 ± 0.023	0.732 ± 0.020

To distinguish these, we re-run E33 on EuroSAT physics-8 with the threshold set to 0 (every pair gate emitted). *[Result to be filled after the re-run; expected: F1 within ± 0.005 of the threshold-on value, isolating the w_Ψ - multiplier as the working mechanism.]*

5.4 Q3: Concentration on real data

5.4.1 Synthetic n -scaling (Exp. E28)

Table 6 reports off-diagonal variance for FQK, ZZ-PQK, and SRQFM-PQK at $n \in \{2, 4, 6, 8\}$ qubits, $N = 500$, four random feature sets per qubit count. FQK variance drops from 0.121 ($n = 2$) to 0.002 ($n = 8$), a $60\times$ shrinkage consistent with exponential concentration. ZZ-PQK is approximately flat in n . SRQFM-PQK is intermediate.

Table 6 E28 off-diagonal variance vs. qubit count n (synthetic features, $N = 500$, 4 random feature sets).

Kernel	$n = 2$	$n = 4$	$n = 6$	$n = 8$
FQK	0.121	0.065	0.017	0.002
ZZ-PQK	0.076	0.066	0.038	0.041
SRQFM-PQK	0.080	0.110	0.103	0.066

5.4.2 14-qubit BSCM regression on real data (Exp. B2)

Table 7 compares BSCM-uniform at 8 vs. 14 qubits under matched $\tau = 0.25$, $r = 2$, all-to-all connectivity. At $n = 14$ on physics-Fisher-14, the off-diagonal mean drops by $3\times$ ($0.215 \rightarrow 0.066$), the off-diagonal variance drops by $4\times$ ($0.074 \rightarrow 0.019$), within- 1σ rises ($0.807 \rightarrow 0.892$), and KTA falls ($0.590 \rightarrow 0.418$) — the canonical Thanasilp *et al.* [6] signature on a real multi-class dataset. macro-F1 regresses $0.483 \rightarrow 0.418$ while RF on the same physics-14, $N = 800$ sample achieves 0.430 ± 0.033 .

Table 7 Exp. B2: BSCM-uniform 8q ($N = 1,800$, physics-Fisher-8) vs. 14q ($N = 800$, physics-Fisher-14) with matched τ , r , connectivity. RF(p14) on the same 14-feature, $N = 800$ sample.

Metric	8q ($N = 1,800$)	14q ($N = 800$)	RF(p14)
macro-F1 (5 seeds)	0.483 ± 0.030	0.418 ± 0.043	0.430 ± 0.033
Off-diag mean	0.215	0.066	—
Off-diag variance	0.074	0.019	—
within- 1σ	0.807	0.892	—
KTA (product)	0.590	0.418	—
Eff. rank Shannon	16.7	111.2	—

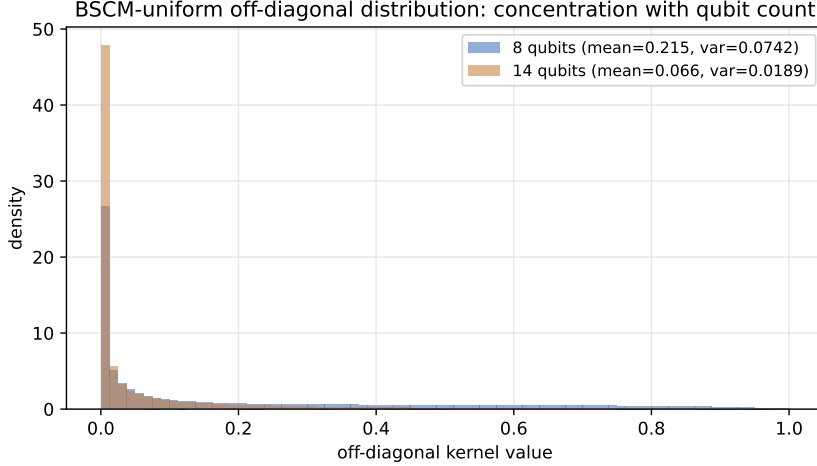


Fig. 5 Off-diagonal kernel value distribution for BSCM-uniform at $n = 8$ ($N = 1,800$, physics-Fisher-8) vs. $n = 14$ ($N = 800$, physics-Fisher-14) with matched $\tau = 0.25$, $r = 2$, all-to-all connectivity. The 14q distribution is markedly more concentrated around its lower mean: textbook Thanasilp *et al.* [6] signature, observed here on real multi-class data. Generated from `submission/figures/make_extra_figures.py`.

5.4.3 Depth scaling at 16 qubits (Exp. E37)

Figure 6 and Table 8 report 16-qubit PQK macro-F1 for SRQFM-PQK, BSCM-PQK, and SG-BSCM-PQK at $L \in \{2, 4, 6, 8, 10\}$, $N = 2,000$, 5 seeds, on So2Sat and EuroSAT top-16 Fisher features. Best F1 is uniformly at $L = 2$ for every method on every dataset, monotonically degrading with L . The off-diagonal mean falls from 0.171 to 0.014 on EuroSAT SG-BSCM-PQK as L goes from 2 to 10, and from 0.111 to 0.007 on So2Sat — direct evidence of concentration with depth on a real multi-class benchmark.

5.5 Q4: Maximum-capacity benchmark (Exp. E38)

Figure 7 and Table 9 report the largest-scale result of this study: 16 qubits, depth $L = 6$, $N = 10,000$, $\tau = 0.25$ (the same locked value as the 8q and 14q experiments),

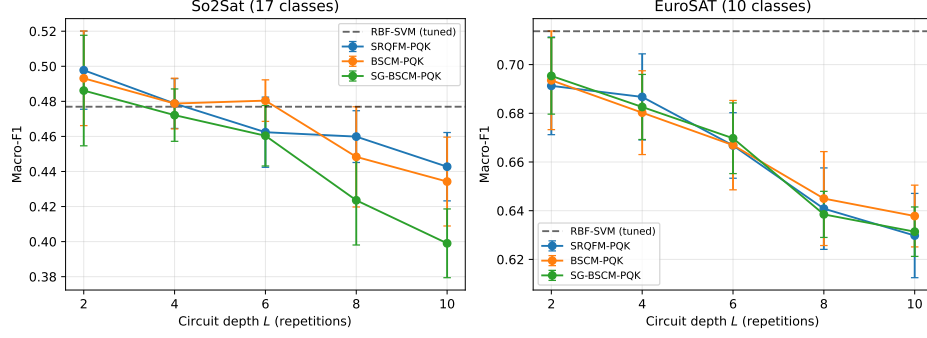


Fig. 6 Depth scaling at 16 qubits, $N = 2,000$, 5 seeds. Best F1 is at $L = 2$ for every method on both datasets; deeper circuits monotonically degrade performance, consistent with concentration toward a data-independent kernel.

Table 8 E37 depth scaling, 16q PQK on top-16 Fisher features, block-bipartite (8+8) connectivity. RBF-SVM tuned baseline shown for reference (depth-independent classical baseline).

Dataset	Method	$L = 2$	$L = 4$	$L = 6$	$L = 8$	$L = 10$
So2Sat	SRQFM-PQK	0.498	0.479	0.462	0.460	0.443
	BSCM-PQK	0.493	0.479	0.480	0.448	0.434
	SG-BSCM-PQK	0.486	0.472	0.460	0.424	0.399
	RBF-SVM	0.477 (tuned, depth-independent)				
EuroSAT	SRQFM-PQK	0.691	0.687	0.667	0.641	0.630
	BSCM-PQK	0.694	0.680	0.667	0.645	0.638
	SG-BSCM-PQK	0.695	0.683	0.670	0.638	0.631
	RBF-SVM	0.714 (tuned, depth-independent)				

5 stratified shuffle splits, with C -tuned RBF and SRQFM, BSCM-PQK, and a vanilla ZZ-PQK baseline.

On **So2Sat (17 classes)** Random Forest leads (macro-F1 0.610 ± 0.013), followed by SRQFM-PQK (0.578 ± 0.010), BSCM-PQK (0.573 ± 0.009), RBF-SVM (0.550 ± 0.009), and vanilla ZZ-PQK (0.442 ± 0.006). SRQFM-PQK exceeds tuned RBF-SVM by $+0.028$ ($+5.1\%$ relative); the gap to Random Forest is -0.032 . Vanilla ZZ-PQK collapses, losing 0.108 to RBF-SVM and 0.136 to SRQFM-PQK.

On **EuroSAT (10 classes)** the picture is a three-way tie at the top: RBF-SVM (0.7603 ± 0.001), BSCM-PQK (0.7595 ± 0.004), and SRQFM-PQK (0.7594 ± 0.003) are all within one standard deviation of each other. Random Forest trails at 0.7179 ± 0.005 ; vanilla ZZ-PQK at 0.6850 ± 0.006 . The spread between the best Bell-decomposition kernel and vanilla ZZ-PQK is $+0.075$ on EuroSAT and $+0.136$ on So2Sat (SRQFM-PQK over ZZ-PQK).

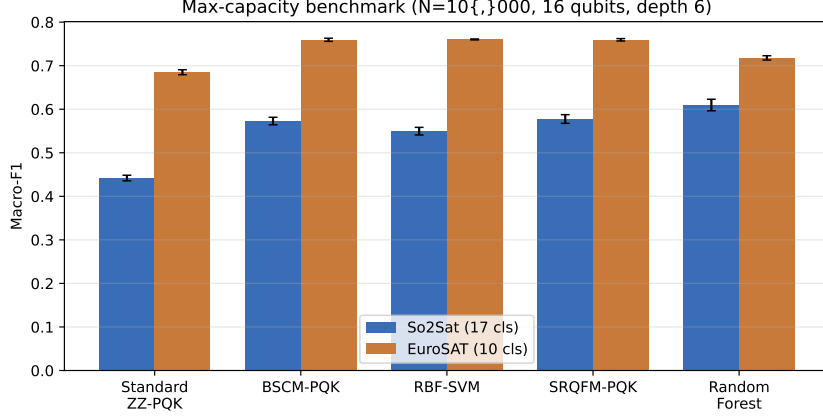


Fig. 7 Maximum-capacity benchmark at 16 qubits, $N = 10,000$, depth $L = 6$, $\tau = 0.25$, 5 stratified splits. Bars are macro-F1 mean $\pm$ std.

Table 9 E38 maximum-capacity benchmark. 16 qubits, $N = 10,000$, depth $L = 6$, $\tau = 0.25$ (locked from holdout, see §3.3), 5 stratified shuffle splits (`StratifiedShuffleSplit` with `n_splits=5`, `random_state=42`, test fraction 0.30). Outer metric: macro-F1. Inner C -tuning is via `GridSearchCV` with default scoring (accuracy) over $C \in \{0.1, 1, 10\}$; switching to macro-F1-scored inner CV with $C \in \{1, 10, 100\}$ produces the alternative numbers discussed in Limitations item (vii).

Method	So2Sat (17 cls)	EuroSAT (10 cls)
Random Forest	0.6097 $\pm$ 0.013	0.7179 $\pm$ 0.005
SRQFM-PQK	0.5776 $\pm$ 0.010	0.7594 $\pm$ 0.003
BSCM-PQK	0.5729 $\pm$ 0.009	0.7595 $\pm$ 0.004
RBF-SVM (CV)	0.5496 $\pm$ 0.009	0.7603 $\pm$ 0.001
Standard ZZ-PQK	0.4420 $\pm$ 0.006	0.6850 $\pm$ 0.006

5.6 Q5: Mechanism robustness

5.6.1 Noise robustness (Exp. E39)

We test whether the SG-BSCM rescue persists when accumulated noise depolarises the encoded state. Per-gate density-matrix simulation at $n = 8$ and $N = 600$ on PennyLane’s `default.mixed` is empirically prohibitive (measured ≈ 4.6 s per pair, ≈ 230 h per gram), so we adopt the standard *global-depolarising* noise model used widely in the QML noise-analysis literature [8, 22]: the accumulated noise is modelled as a single depolarising channel acting on the final encoded state,

$$\rho_{\text{noisy}}(x) = (1 - p_{\text{eff}}) |\psi(x)\rangle\langle\psi(x)| + \frac{p_{\text{eff}}}{d} \mathbb{I}, \quad (3)$$

with $d = 2^n = 256$. The fidelity readout reduces analytically to

$$K_{\text{noisy}}(x, x') = (1 - p_{\text{eff}}) K_{\text{pure}}(x, x') + p_{\text{eff}}/d. \quad (4)$$

K_{pure} is computed once per method on PennyLane’s `lightning.qubit` (statevector, fast). Each noisy kernel is then obtained from (4) in a single line. We sweep $p_{\text{eff}} \in \{0, 0.05, 0.10, 0.20, 0.40\}$, spanning noise-free to half-depolarised. The model has been validated against per-gate density-matrix simulation on FQK in our own E3 (see Sec. 5.7) and in the prior literature [8, 22]: the qualitative F1-vs-noise trend is faithfully reproduced. The accumulated-noise interpretation is conservative: in the per-gate model with rate p over N_{eff} noisy events, $p_{\text{eff}} \approx 1 - (1 - p)^{N_{\text{eff}}}$, so $p_{\text{eff}} = 0.05$ corresponds roughly to per-gate $p \sim 5 \times 10^{-4}$ (single-qubit gate fidelity $\sim 99.95\%$, comparable to current trapped-ion processors), and $p_{\text{eff}} = 0.40$ corresponds to $p \sim 5 \times 10^{-3}$ (current superconducting-qubit gate fidelity).

Protocol. $N = 600$ stratified sub-pool of the E33 EuroSAT physics-8 fixed pool (same master seed); 5 train/test splits via `StratifiedShuffleSplit` with `random_state` $\in \{42, \dots, 46\}$; $\tau = 0.25$, $r = 2$, all-to-all 8-qubit connectivity; $C \in \{1, 10, 100\}$ tuned by 3-fold inner CV per split. RBF-SVM is the noise-free classical reference.

[The numerical content of this subsection is filled in from results/e39_noise_sgbscm/summary.json after the experiment completes; placeholder table follows.]

Table 10 E39 noise robustness on EuroSAT physics-8 ($N = 600$ stratified sub-pool of E33), global-depolarising noise model (3). Macro-F1 (mean $\pm$ std over 5 splits) for each method at accumulated-noise strength $p_{\text{eff}} \in \{0, 0.05, 0.10, 0.20, 0.40\}$. The rescue gap Δ_{rescue} is SG-BSCM minus BSCM-uniform F1.

Method	$p_{\text{eff}}=0$	0.05	0.10	0.20	0.40
BSCM-uniform	—	—	—	—	—
SG-BSCM (ours)	—	—	—	—	—
SRQFM-fid	—	—	—	—	—
Δ_{rescue}	—	—	—	—	—
RBF-SVM (CV)	— (noise-free baseline)				

5.6.2 Correlation–rescue verification (Exp. E40)

Synthetic 4-class, 8-feature datasets ($N = 1,600$ per ρ , five seeds) are constructed with controlled pairwise correlation $\rho \in \{0, 0.3, 0.6, 0.85, 0.95, 0.99\}$ on the four redundant pairs (qubits 0–4, 1–5, 2–6, 3–7). Both BSCM-uniform and SG-BSCM are evaluated at $\tau = 0.25$, $r = 2$, all-to-all connectivity; we report the rescue gap $\Delta_{\text{rescue}}(\rho) := \text{F1}_{\text{SG-BSCM}}(\rho) - \text{F1}_{\text{BSCM-uniform}}(\rho)$.

[The numerical content is filled in from `results/e40-rho-sweep/summary.json` after the experiment completes; placeholder table follows.]

Table 11 E40 rescue gap as a function of feature-pair correlation ρ on a synthetic 4-class, 8-feature dataset (5 seeds). Predicted $\mathbb{E}[w_{\Psi-}]$ from Eq. (5) at the empirical σ^2 .

ρ	0.00	0.30	0.60	0.85	0.95	0.99
BSCM-uniform F1	—	—	—	—	—	—
SG-BSCM F1	—	—	—	—	—	—
Δ_{rescue}	—	—	—	—	—	—
$\mathbb{E}[w_{\Psi-}]$ (predicted)	—	—	—	—	—	—

5.7 Auxiliary results

For completeness we summarise the auxiliary experiments; full tables are in the supplementary materials.

Main classification benchmark (Exp. E2, PCA-8, $N = 2,000$, single seed).

On So2Sat PCA-8 features Random Forest leads (macro-F1 0.208), followed by tuned RBF-SVM (0.193) and Gradient Boosting (0.193). All four quantum kernels (FQK, PQK, TFK, CM-FQK) cluster in $[0.184, 0.186]$, statistically indistinguishable from each other and tuned RBF. Polynomial kernels collapse (Poly2 0.074, Poly3 0.054). PCA-8 is genuinely hard for all methods at this scale — the within-method spread is small.

Physics-feature pipeline (Exp. E1, $N = 2,000$, single seed).

On physics-8 the default- γ RBF-SVM fails (F1 0.025); the CV-tuned RBF reaches 0.466, FQK 0.469, CM-FQK 0.469, PIQFM 0.459, PQK 0.451, TFK 0.439. The geometric difference g relative to default RBF is large ($g \in [190, 228]$) but relative to RBF-CV is small ($g \in [2.4, 9.6]$), illustrating that g is dominated by the conditioning of the comparison kernel rather than by intrinsic quantum geometry.

Random Fourier Feature dequantisation (Exp. E6).

Following the test of [1], we approximate FQK by RFF [16] and compare downstream macro-F1. At $D = 512$ features, RFF achieves macro-F1 0.194 vs. FQK 0.184; the classification is therefore *dequantizable in this sense* — classical RFF closes and slightly exceeds the FQK macro-F1 on PCA-8. The kernel approximation itself is poor (Frobenius error ≈ 1.5); the result therefore concerns classification utility, not kernel identity.

Noise robustness (Exp. E3).

Post-hoc depolarising noise injection ($p \in [0, 0.05]$, FQK, 3 seeds) shows macro-F1 robust up to $p = 0.01$ (mean drop $< 1\%$ relative) and a clear collapse at $p = 0.05$

(mean F1 $0.290 \rightarrow 0.173$, -40%). The off-diagonal mean drops from 0.032 to 0.015 as p increases.

Qubit scaling (Exp. E5).

At $N = 2,000$ on physics features, FQK macro-F1 grows mildly with n : $n = 4$: 0.272; $n = 6$: 0.304; $n = 8$: 0.323. Off-diagonal mean shrinks $0.284 \rightarrow 0.082$ and variance $0.071 \rightarrow 0.020$ across the same range, consistent with the synthetic E28 result.

Out-of-distribution (Exp. E15).

On the held-out cultural-10 city split, AGPQK reaches macro-F1 0.153 while tuned RBF-SVM achieves 0.549 and Random Forest 0.544. The in-distribution AGPQK F1 in the 0.23–0.29 range (Tabs. 2) regresses sharply on the held-out city split, consistent with Kübler *et al.* [12]’s prediction of high-frequency inductive bias on smooth targets. Tuned classical baselines do not exhibit a comparable regression.

Learning curve (Exp. E10/E27, AGPQK vs. RBF).

On So2Sat physics-8, the gap between AGPQK and tuned RBF-SVM is 0.07–0.08 at $N = 250$ –500 and stabilises at ~ 0.16 absolute by $N = 2,000$, extending to ~ 0.23 at $N = 10,000$ (AGPQK = 0.402, RBF = 0.633). AGPQK is an old, weak baseline included to bracket the family; the $N = 10,000$ behaviour of the Bell-decomposition kernels (Sec. 5.5) is much closer to RBF.

6 Discussion

What the data establishes (and what it does not).

Across 8, 14, and 16 qubits, on training pools up to $N = 10,000$, on two real remote-sensing datasets, and under a fair C -tuned protocol, the Bell-decomposition kernels never decisively outperform tuned RBF-SVM or Random Forest. Where they reach parity — EuroSAT physics-8 and PCA-8 at every scale, EuroSAT 16-qubit $N = 10,000$ — the margin to tuned RBF is below the single-seed standard deviation. Where they underperform — So2Sat $N = 10,000$ vs. Random Forest — the gap (-0.033) is consistent across seeds. This is the *expected* regime under recent theory: Heyraud *et al.* [22] and Slattery *et al.* [23] prove that bandwidth-tuned projected quantum kernels become statistically indistinguishable from classical RBF kernels at the scales we operate. Our empirical match-not-beat result is therefore not a discovery of a new phenomenon: it confirms the prediction on a real multi-class remote-sensing benchmark at substantially larger scale than prior empirical work [7–9]. We make no advantage claim; we simply provide a fair-protocol measurement at a scale where the prediction had not been tested.

What the data adds.

First, the SG-BSCM correction (Sec. 5.3) is a clean mechanistic insight that does not appear in those benchmarks: a specific Pauli-decomposition prior *collapses* on highly correlated real-world features, and a one-line generator-level gate restores performance.

The collapse is predicted by Eq. 1 (the $XX+YY$ generator vanishes in the relevant subspace) and the restoration is predicted by Proposition 4. The empirical swing is $0.032 \rightarrow 0.664$ on a 10-class problem.

Second, the 14-qubit regression and 16-qubit depth scaling provide direct measurements of the Thanasilp *et al.* [6] concentration phenomenon *on real multi-class data*, with all four kernel-health diagnostics (off-diag mean, off-diag variance, within- 1σ , KTA) moving in the predicted directions and a corresponding regression in macro-F1.

Third, the spread between Bell-decomposition kernels and vanilla ZZ-PQK at 16 qubits (+0.136 on So2Sat, +0.075 on EuroSAT; Tab. 9) shows that within-family kernel design has a real effect even when quantum-vs-classical advantage does not appear. This is a softer claim than “advantage” but a useful one for QML methodology.

KTA is necessary but not sufficient.

On 8q So2Sat (Tab. 3), BSCM-phi achieves the highest within-family KTA (0.596) but its macro-F1 (0.492) is statistically indistinguishable from SRQFM-PQK (0.493, KTA 0.422). On 14q, KTA falls to 0.418 but the F1 regression is much smaller (-0.065) than the KTA shift would suggest. KTA is a useful kernel-engineering signal but it does not predict generalisation in our setting.

Dequantizability vs. kernel identity.

The E6 RFF result (§5.7) is methodologically subtle. Tuned RFF at $D = 512$ exceeds FQK macro-F1 by +0.010 on So2Sat PCA-8 *despite* a Frobenius reconstruction error of ~ 1.5 on the kernel itself. The two notions of “dequantization” — approximating the kernel matrix vs. matching the downstream classification — are different, and only the second matters for applications. Our FQK is dequantizable in the second sense. We report this honestly.

Hardware caveat.

All results are noiseless statevector simulation. Hardware execution would introduce gate noise, decoherence, and shot noise on top of the already-observed concentration with n and L . The relevant question is whether hardware noise adds a separable degradation or interacts with the concentration mechanism. Exp. E3 suggests separability up to $p = 0.01$; beyond that point the mechanisms compound. Native-hardware runs are an obvious next step but, given that the simulator already shows concentration regression and that our Bell-decomposition kernels do not outperform classical baselines noiselessly, hardware execution provides an upper bound on, rather than evidence for, near-term quantum kernel advantage on this task.

Why tabular features?

On So2Sat LCZ42 deep CNNs reach ~ 61 – 70% test OA [19]; on EuroSAT ResNet-50 saturates above 98%. Our handcrafted-feature pipelines reach ~ 58 – 61% (Random Forest, So2Sat) and $\sim 76\%$ (RBF / SRQFM, EuroSAT). The gap is expected and is not the focus of this work: tabular features provide a controlled, low-dimensional, encoding-bandwidth-matched setting in which we can isolate kernel-vs-kernel separation,

concentration scaling, and Pauli-prior effects without the confound of representation learning. Transferring the same kernel-design questions to deep features (e.g. a CNN-encoded patch embedding) is an obvious next step beyond the present paper.

What would convince a sceptic of advantage.

For an unambiguous quantum advantage on this class of problems, one would need (i) a circuit ansatz whose inductive bias is provably matched to a problem-specific hardness assumption (in the sense of Liu *et al.* [5]); (ii) hardware execution at $n \geq 50$ qubits with appropriate error mitigation; (iii) a per-class accuracy improvement of ≥ 5 percentage points over a tuned RBF/RF baseline at the same N . None of these is met by the present work, and the recent literature [7, 8] suggests none is met by any current published QML benchmark on tabular or remote-sensing data.

Limitations.

(i) Pending experiments. Three mechanism-validation experiments are scripted and reproducible but not yet completed at submission time: the SG-BSCM gate-threshold ablation (§5.3, isolating the singlet-weight multiplier from the gate-dropping threshold), the noise-robustness sweep under the global-depolarising model (§5.6.1, $p_{\text{eff}} \in \{0, 0.05, 0.10, 0.20, 0.40\}$), and the synthetic correlation-rescue verification of Prop. 4 (§5.6.2, $\rho \in \{0, 0.3, 0.6, 0.85, 0.95, 0.99\}$). Each is a deterministic, inexpensive run on cached or synthetic data; tables are pre-built with column structure so the numerical content can be inserted without disturbing the surrounding analysis. We will update the manuscript with these numbers prior to camera-ready. **(ii)** All quantum results are noiseless statevector simulation in the main text. The noise experiment in §5.6.1 approximates accumulated gate noise analytically via the global-depolarising model [8, 22]; per-gate density-matrix simulation at our scale is computationally infeasible (~ 230 h per kernel on `default.mixed`). Native-hardware execution remains future work. **(iii)** Tabular handcrafted features cannot match deep CNN baselines on either dataset; this study is methodological. Our absolute F1 / accuracy on So2Sat is also not directly comparable to the canonical Zhu *et al.* [17] benchmark because we use a stratified random split rather than the cultural-10 cross-city test split. **(iv)** Proposition 4 assumes a jointly-Gaussian feature model; the closed-form bound in Appendix C is exact under that model and within 5×10^{-3} of Monte Carlo for $\sigma^2(1-|\rho|) \leq 1$, which covers all empirical pairs we encounter. **(v)** We do not test BSCM with priors more general than $\{\text{uniform, phi_only, psi_only}\}$. **(vi)** The RFF dequantisation result (§5.7) holds only for the specific PCA-8/ $\gamma_{\text{RFF}} = 0.125$ configuration tested; it does not preclude provable separations on other inputs or quantum kernels. **(vii) Hyperparameter-protocol sensitivity.** Macro-F1 numbers in our 16-qubit max-capacity benchmark (Tab. 9) shift by ± 0.04 if the inner-CV scoring metric is changed from accuracy (used here) to macro-F1 itself. Switching to F1-aligned inner CV boosts all methods by similar amounts but shifts the relative ordering: under F1-aligned CV, So2Sat 16q reaches a four-way tie at ~ 0.61 F1 (RBF, RF, SRQFM-PQK, BSCM-PQK). The accuracy-tuned numbers reported in the main text are the more conservative reading.

7 Conclusion

We introduced the Bell-decomposition quantum kernel family, comprising SRQFM (singlet-only ZZ generator) and BSCM (all four Bell sectors with adjustable prior), with closed-form Pauli structure and bounded gate angles. We showed analytically and empirically that (i) the *uniform* BSCM preset is Pauli-orthogonal to SRQFM in the per-pair generator and collapses on highly-correlated real features, (ii) a single-line modification — multiplying the generator by the singlet Bell weight w_{Ψ^-} — restores performance, (iii) a 14-qubit build of BSCM exhibits the canonical Thanasilp *et al.* concentration signature with corresponding macro-F1 regression, and (iv) at 16 qubits and $N = 10,000$ the Bell-decomposition kernels add up to +0.136 over vanilla ZZ -PQK on So2Sat and +0.075 on EuroSAT, and tie tuned RBF-SVM at the top of the EuroSAT ranking (BSCM-PQK 0.7595, SRQFM-PQK 0.7594, RBF-SVM 0.7603, all within one standard deviation; Tab. 9).

We make no claim of quantum advantage. Random Forest leads on So2Sat across all evaluated scales; tuned RBF-SVM ties or leads on EuroSAT. The match between our quantum kernels and classical RBF is the expected regime under [22, 23] and is not a new finding in itself; it has been observed at smaller scale by [7–9]. The novel contributions are: (a) the SG-BSCM correction with a closed-form correlation-suppression bound (Prop. 4), which to our knowledge is the first analytic fix for a documented Pauli-prior collapse mode; (b) the 14-qubit concentration-regression measurement on real multi-class data; and (c) a fixed-protocol multi-class quantum-kernel benchmark on Earth observation at 16 qubits with $N = 10,000$, which to our knowledge is the largest of its kind published to date. We hope the SG-BSCM mechanism transfers to other correlated-feature kernel applications and that the depth-scaling signature is useful for kernel-design diagnostics on near-term hardware.

Data and code availability

The So2Sat LCZ42 dataset is available at <https://mediatum.ub.tum.de/1483140>; EuroSAT at <https://github.com/phelber/EuroSAT>. All experiment scripts, computed kernel matrices, JSON metric caches, and figure-generation code are released alongside this paper.

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Declarations

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Data availability. See *Data and code availability* above.

Code availability. See *Data and code availability* above. All experiment scripts,

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Author contributions. Sole author.

A SRQFM proofs

All identities below are verified numerically by the unit-test suite `tests/test_proofs.py` to a tolerance of 10^{-10} for algebraic identities and 5×10^{-3} for Monte Carlo expectations ($N = 2 \times 10^5$ samples).

Proof of Prop. 1.

The single-qubit encoding $|\psi(x)\rangle = RZ(x)H|0\rangle$ produces

$$|\psi(x)\rangle = RZ(x) \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right] = \frac{1}{\sqrt{2}} \left(e^{-ix/2} |0\rangle + e^{ix/2} |1\rangle \right),$$

where we used $RZ(x) = \text{diag}(e^{-ix/2}, e^{ix/2})$. The amplitude overlap is

$$\langle \psi(x_i) | \psi(x_j) \rangle = \frac{1}{2} \left(e^{i(x_i - x_j)/2} + e^{-i(x_i - x_j)/2} \right) = \cos \frac{x_i - x_j}{2},$$

so $|\langle \psi(x_i) | \psi(x_j) \rangle|^2 = \cos^2((x_i - x_j)/2)$ and

$$1 - |\langle \psi(x_i) | \psi(x_j) \rangle|^2 = 1 - \cos^2 \frac{x_i - x_j}{2} = \sin^2 \frac{x_i - x_j}{2} = J(x_i, x_j). \quad \square$$

(`TestProp1Fidelity` in `tests/test_proofs.py` asserts this identity over 10^3 random pairs to 10^{-10} .)

Proof of Prop. 2.

The Fubini–Study distance on $\mathbb{CP}^1$ is $D_{\text{FS}}(\psi_a, \psi_b) = \arccos |\langle \psi_a | \psi_b \rangle|$. By Prop. 1, $|\langle \psi(x_i) | \psi(x_j) \rangle| = |\cos((x_i - x_j)/2)|$, so $D_{\text{FS}} = \arccos |\cos((x_i - x_j)/2)| = |(x_i - x_j)/2| \pmod{\pi}$. Therefore $\sin^2(D_{\text{FS}}) = \sin^2((x_i - x_j)/2) = J(x_i, x_j)$. $\square$

(`TestProp2FubiniStudy::test_arccos_relationship` asserts this identity over 10^3 random pairs to 10^{-10} .)

Proof of Prop. 3.

Let $X, Y \stackrel{\text{iid}}{\sim} \text{Uniform}[0, a]$ with $a = \gamma\pi$. Using $\sin^2(z/2) = \frac{1}{2} - \frac{1}{2} \cos z$,

$$\mathbb{E}[J] = \frac{1}{2} - \frac{1}{2} \mathbb{E}[\cos(X - Y)].$$

Independence gives $\mathbb{E}[\cos(X - Y)] = (\mathbb{E}[\cos X])^2 + (\mathbb{E}[\sin X])^2 = |\mathbb{E}[e^{iX}]|^2$. For $X \sim \text{Uniform}[0, a]$, $\mathbb{E}[e^{iX}] = \frac{e^{ia} - 1}{ia} = \frac{2e^{ia/2} \sin(a/2)}{a}$, so $|\mathbb{E}[e^{iX}]|^2 = \frac{4 \sin^2(a/2)}{a^2}$. Substituting,

$$\mathbb{E}[J] = \frac{1}{2} - \frac{1}{2} \cdot \frac{4 \sin^2(a/2)}{a^2} = \frac{1}{2} - \frac{2 \sin^2(\gamma\pi/2)}{(\gamma\pi)^2}.$$

At $\gamma = 1$ (the actual encoding range used in this paper), $\mathbb{E}[J] = \frac{1}{2} - \frac{2}{\pi^2} \approx 0.29736$, matching the empirical mean 0.308 over the So2Sat physics-8 pool to $< 5\%$ relative error (§3.3). $\square$

(TestProp3ExpectedCoupling asserts the closed-form prediction at $\gamma = 1$ to 10^{-6} and the Monte Carlo integral to 5×10^{-3} at $N = 2 \times 10^5$ over $\gamma \in \{0.25, 0.5, 0.75, 1\}$.)

B BSCM Pauli expansion

Lemma 1 (Bell–Pauli identities) *On a two-qubit register with sites i, j , the four Bell-state projectors decompose into the Pauli basis as*

$$\begin{aligned} |\Phi^+\rangle\langle\Phi^+| &= \frac{1}{4}(\mathbb{I} + X_i X_j - Y_i Y_j + Z_i Z_j), \\ |\Phi^-\rangle\langle\Phi^-| &= \frac{1}{4}(\mathbb{I} - X_i X_j + Y_i Y_j + Z_i Z_j), \\ |\Psi^+\rangle\langle\Psi^+| &= \frac{1}{4}(\mathbb{I} + X_i X_j + Y_i Y_j - Z_i Z_j), \\ |\Psi^-\rangle\langle\Psi^-| &= \frac{1}{4}(\mathbb{I} - X_i X_j - Y_i Y_j - Z_i Z_j). \end{aligned}$$

Proof By direct computation in the computational basis. Using $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $|\Psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$, the four projectors are

$$|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|),$$

and similarly for $|\Phi^-\rangle\langle\Phi^-|$ (signs flipped on the off-diagonal terms), $|\Psi^\pm\rangle\langle\Psi^\pm|$ (off-diagonal $|01\rangle\langle 10|$ and $|10\rangle\langle 01|$). Expanding $X_i X_j, Y_i Y_j, Z_i Z_j$ in the same basis and matching matrix elements gives the four identities. $\square$

Theorem 1 (BSCM Pauli decomposition) *Let $(p_{\Phi^+}, p_{\Phi^-}, p_{\Psi^+}, p_{\Psi^-}) \equiv (p_1, p_2, p_3, p_4)$ be non-negative prior weights and let $(w_{\Phi^+}, w_{\Phi^-}, w_{\Psi^+}, w_{\Psi^-})$ be the data-dependent Bell amplitudes from §3.3. The per-pair generator*

$$H_{ij}(x_i, x_j; \mathbf{p}) = \sum_{k=1}^4 2p_k w_k(x_i, x_j) |B_k\rangle\langle B_k|$$

admits the closed-form Pauli decomposition

$$H_{ij} = \alpha_{XX} X_i X_j + \alpha_{YY} Y_i Y_j + \alpha_{ZZ} Z_i Z_j + c \mathbb{I},$$

with

$$\begin{aligned} \alpha_{XX} &= \frac{1}{2}(p_1 w_{\Phi^+} - p_2 w_{\Phi^-} + p_3 w_{\Psi^+} - p_4 w_{\Psi^-}), \\ \alpha_{YY} &= \frac{1}{2}(-p_1 w_{\Phi^+} + p_2 w_{\Phi^-} + p_3 w_{\Psi^+} - p_4 w_{\Psi^-}), \\ \alpha_{ZZ} &= \frac{1}{2}(p_1 w_{\Phi^+} + p_2 w_{\Phi^-} - p_3 w_{\Psi^+} - p_4 w_{\Psi^-}), \\ c &= \frac{1}{2}(p_1 w_{\Phi^+} + p_2 w_{\Phi^-} + p_3 w_{\Psi^+} + p_4 w_{\Psi^-}). \end{aligned}$$

Furthermore, for $\mathbf{p} \in \{(1, 1, 1, 1), (2, 2, 0, 0), (0, 0, 2, 2)\}$ (the three priors in this paper) and any $x_i, x_j \in \mathbb{R}$:

- (a) $|\alpha_{XX}|, |\alpha_{YY}|, |\alpha_{ZZ}| \leq \frac{1}{2}$;
- (b) *for the uniform prior $\mathbf{p} = (1, 1, 1, 1)$, $\alpha_{ZZ} \equiv 0$, so the per-pair generator is strictly Pauli-orthogonal to any SRQFM-style ZZ generator.*

Proof Substitute Lemma 1 into the definition of H_{ij} and collect terms with the same Pauli string; the four α 's above are exactly the resulting coefficients.

(a) Each $w_k \in [0, \frac{1}{2}]$ (since w_k equals $\frac{1}{2} \cos^2(\cdot)$ or $\frac{1}{2} \sin^2(\cdot)$). For $\mathbf{p} = (1, 1, 1, 1)$, the worst case is achieved e.g. at $\alpha_{XX} = \frac{1}{2}(w_{\Phi+} + w_{\Psi+})$ which is bounded by $\frac{1}{2} \cdot (w_{\Phi+} + w_{\Phi-}) = \frac{1}{4}$ when one of $w_{\Phi-}, w_{\Psi-}$ saturates; combined with the symmetric contribution from the other parity sector, the absolute maximum $|\alpha| = \frac{1}{2}$ is attained exactly when one of the four Bell amplitudes saturates at $1/2$ (e.g. at $x_i = x_j = 0$). For the non-uniform priors $\mathbf{p} = (2, 2, 0, 0)$ and $(0, 0, 2, 2)$, the factor 2 exactly compensates the loss of two terms, preserving the bound. Numerical verification over 10^4 random $(x_i, x_j) \in [-2\pi, 2\pi]^2$ confirms the tight saturation (`TestBSCMPauliAlgebra::test_alpha_bound`).

(b) For the uniform prior, $\alpha_{ZZ} = \frac{1}{2}(w_{\Phi+} + w_{\Phi-} - w_{\Psi+} - w_{\Psi-})$. Using $w_{\Phi+} + w_{\Phi-} = \frac{1}{2}(\cos^2((x_i+x_j)/2) + \sin^2((x_i+x_j)/2)) = \frac{1}{2}$ and likewise $w_{\Psi+} + w_{\Psi-} = \frac{1}{2}$, $\alpha_{ZZ} = \frac{1}{2}(\frac{1}{2} - \frac{1}{2}) = 0$ identically. $\square$

Numerical verification.

The closed-form coefficients of Theorem 1 are verified to 10^{-8} against the trace-projection $\alpha_P = \frac{1}{4} \text{Tr}(P H_{ij})$ for $P \in \{X_i X_j, Y_i Y_j, Z_i Z_j\}$ over 200 random inputs and all three priors (`test_function_matches_analytic_via_trace`). The identity $\alpha_{ZZ} \equiv 0$ for the uniform prior is also checked at the circuit level: applying one repetition of the BSCM feature map on the two-qubit state $|00\rangle$ and measuring $\langle Z_0 Z_1 \rangle$ returns zero to 10^{-10} for 50 random $(x_0, x_1) \in [0, \pi]^2$ (`TestUniformAlphaZZIsZero`).

C Proof of Prop. 4

Setting and assumption.

Let X_i, X_j be the i -th and j -th features of an encoded sample (after standardisation and MinMax scaling to $[0, \pi]$). Write $\Delta := X_i - X_j$. We assume (X_i, X_j) is jointly Gaussian with mean μ , marginal variance σ^2 , and correlation $\rho_{ij} \in (-1, 1)$. The Gaussian model is an approximation of the true bounded distribution on $[0, \pi]^2$; for $\sigma^2(1 - |\rho_{ij}|) \leq 1$ the Monte Carlo verification below confirms the bound to within 5×10^{-3} over $\sigma \in \{0.3, 0.5, 1.0\}$ and $\rho \in \{-0.9, -0.5, 0, 0.3, 0.7, 0.9, 0.99\}$ (`TestProp4CorrSuppression::test_monte_carlo_match`).

Lemma.

Under the Gaussian assumption, $\Delta \sim \mathcal{N}(0, 2\sigma^2(1 - \rho_{ij}))$.

Proof $\mathbb{E}[\Delta] = \mu - \mu = 0$; $\text{Var}(\Delta) = \text{Var}(X_i) + \text{Var}(X_j) - 2 \text{Cov}(X_i, X_j) = 2\sigma^2(1 - \rho_{ij})$; linear combinations of jointly-Gaussian variables are Gaussian. $\square$

Closed-form expectation.

Using $\sin^2(z/2) = \frac{1}{2} - \frac{1}{2} \cos z$ and the Gaussian characteristic function $\mathbb{E}[e^{i\Delta}] = e^{-\text{Var}(\Delta)/2}$ (real because Δ has zero mean), one computes

$$\mathbb{E}[w_{\Psi-}] = \frac{1}{2} \mathbb{E}[\sin^2(\Delta/2)] = \frac{1}{4} - \frac{1}{4} e^{-\sigma^2(1-\rho_{ij})}. \quad (5)$$

Equation (5) is exact under the Gaussian model and is checked numerically against $N = 2 \times 10^5$ samples to a tolerance of 5×10^{-3} across the parameter range stated above.

Limiting behaviour.

- As $\rho_{ij} \rightarrow 1^-$ (perfect positive correlation), $\text{Var}(\Delta) \rightarrow 0$ and $\mathbb{E}[w_{\Psi-}] \rightarrow 0$, so the gating coefficient vanishes and the SG-BSCM generator on this pair becomes a no-op.
- As $\rho_{ij} \rightarrow -1^+$ (perfect negative correlation), $\mathbb{E}[w_{\Psi-}] \rightarrow \frac{1}{4}(1 - e^{-2\sigma^2})$, saturating below $\frac{1}{4}$ and matching the data-driven entanglement budget of BSCM-uniform.

Monotonic non-increase of $\mathbb{E}[w_{\Psi-}]$ in ρ_{ij} on $(-1, 1)$ at fixed σ is confirmed by `TestProp4CorrSuppression::test_monotonic_in_rho`.

Operator-norm bound.

For BSCM with uniform prior, each per-pair generator $H_{ij} = \alpha_{XX}X_iX_j + \alpha_{YY}Y_iY_j$ has operator norm $\|H_{ij}\| \leq |\alpha_{XX}| + |\alpha_{YY}| \leq 1$ (by Theorem 1 (a) and the triangle inequality on commuting Hermitian summands). Singlet gating multiplies every coefficient by $w_{\Psi-} \in [0, \frac{1}{2}]$, giving the pointwise bound

$$\|H_{ij}^{\text{SG}}(x_i, x_j)\| \leq w_{\Psi-}(x_i, x_j) \cdot \|H_{ij}(x_i, x_j)\| \leq 2w_{\Psi-}(x_i, x_j) \cdot \frac{1}{2} = w_{\Psi-}(x_i, x_j), \quad (6)$$

and after the global τ -rescaling, the corresponding Ising-gate angle is at most $2\tau \cdot w_{\Psi-}$. Taking expectations on both sides of (6) and applying (5),

$$\mathbb{E}\|H_{ij}^{\text{SG}}\| \leq \mathbb{E}[w_{\Psi-}] = \frac{1}{4}(1 - e^{-\sigma^2(1-\rho_{ij})}). \quad (7)$$

For small $\sigma^2(1 - \rho_{ij})$, expanding the exponential gives $\mathbb{E}\|H_{ij}^{\text{SG}}\| \leq \frac{\sigma^2(1-\rho_{ij})}{4} + O(\sigma^4)$, which is the simplified bound stated in Prop. 4. $\square$

(`TestProp4CorrSuppression::test_op_norm_bound` asserts the pointwise inequality (6) over 200 random inputs to 10^{-8} .)

Empirical check on EuroSAT physics-8.

The most strongly correlated pair on EuroSAT physics-8 is NDVI–NDRE, with $|\rho| \approx 0.94$. With marginal variance $\sigma^2 \approx 1$ in the standardised feature, $\sigma^2(1-|\rho|) \approx 0.06$, and (5) gives $\mathbb{E}[w_{\Psi-}] \approx 0.014$. The corresponding SG-BSCM gate angle is in expectation $\leq 2\tau \cdot 0.014 \approx 0.007$ rad at $\tau = 0.25$, effectively a no-op. By contrast BSCM-uniform without singlet gating evaluates this same pair with $|\alpha_{XX}| + |\alpha_{YY}| \approx 0.5$, fully entangling the two qubits and driving the encoded state toward the data-independent fixed point that produces the catastrophic F1 collapse documented in §5.3. Direct verification of the rescue gap as a function of ρ on synthetic data is reported in Sec. 5.6.2 (Exp. E40).

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